IoT Laboratory Report

Interfacing Arduino with Node-RED to Monitor Temperature and Humidity

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Hardware	Arduino Uno R3, DHT11 Sensor, Jumper Wires, USB A-B Cable
Software	Arduino IDE, Node.js, Node-RED

Objective

This experiment demonstrates the interfacing of an Arduino Uno R3 with Node-RED using a DHT11 temperature and humidity sensor. Sensor data is transmitted via serial communication, processed by a Node-RED function node, and displayed as live gauge widgets on a web-based dashboard.

Required Node Packages

Install the following packages before building the Node-RED flow:

Package	Purpose	Install Command
`node-red-dashboard`	Gauge & UI widgets for browser dashboard	`npm install node-red-dashboard`
`node-red-node-serialport`	Serial In/Out nodes to read from Arduino's COM port	`npm install node-red-node-serialport`

Alternatively: Node-RED Menu → Manage Palette → Install tab → search and install each package.

Step 1 — Arduino Setup

1.1 Hardware Connection

Connect the DHT11 sensor to the Arduino Uno R3 as follows:

DHT11 Pin	Arduino Pin
VCC	5V or 3.3V
GND	GND
DATA	Digital Pin 2

1.2 Arduino Code

Install the DHT sensor library by Adafruit from the Arduino IDE Library Manager (Sketch → Include Library → Manage Libraries). Then upload the following code:

```
#include <DHT.h>
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  dht.begin();
  Serial.begin(9600);
}

void loop() {
  float H = dht.readHumidity();
  float T = dht.readTemperature();
  if (isnan(H) || isnan(T)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
  Serial.print("Humidity:");
  Serial.print(H);
  Serial.print(",Temp:");
  Serial.println(T);
  delay(2000);
}
```

The code reads humidity and temperature every 2 seconds and sends them over serial in the format:
`Humidity:66.00,Temp:33.80`

Step 2 — Node-RED Flow Setup

Open Node-RED at http://127.0.0.1:1880 and create a new flow with the following nodes.

2.1 Nodes Required in the Flow

Node	Configuration
Serial In	Port: COM6 (or your port), Baud rate: 9600. Named 'Arduino Uno R3'.

Node	Configuration
Function	Parses the serial string and outputs temperature and humidity as separate messages.
Gauge (×2)	One for Temperature (0–50 °C) and one for Humidity (0–100%).
Debug	Connected to function output to verify parsed values in the debug panel.

2.2 Node-RED Flow Diagram

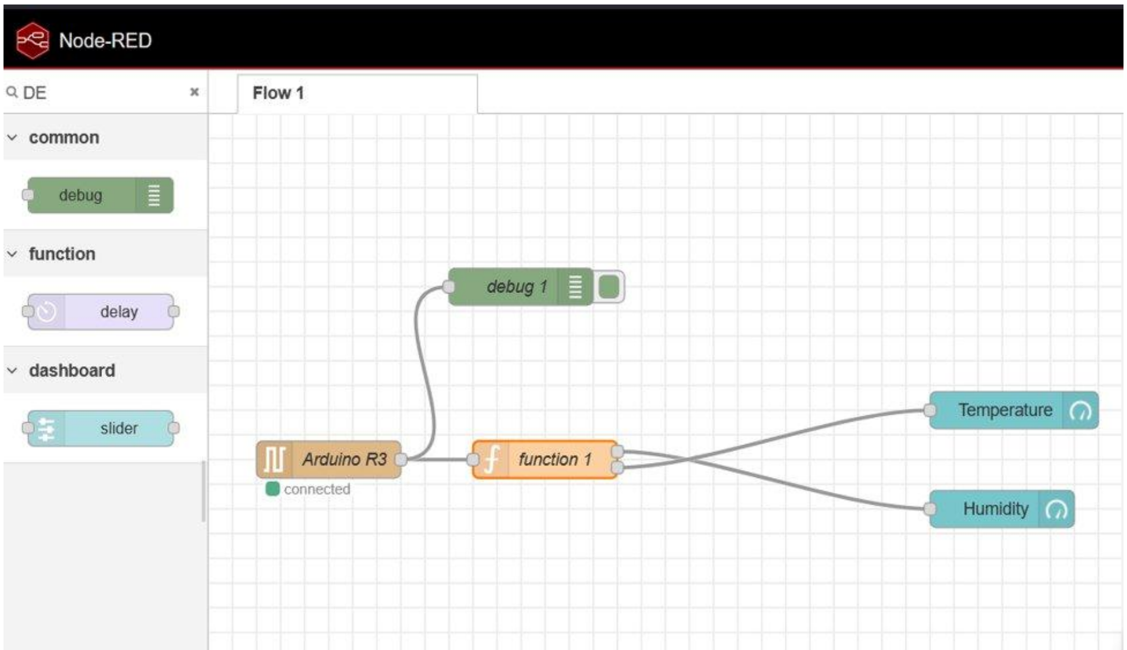

Figure 1: Node-RED Flow — Arduino R3 → Function → Temperature / Humidity Gauges + Debug

2.3 Function Node Code

Double-click the Function node and paste the following code in the 'On Message' tab. The function splits the incoming serial string and routes Temperature (output 1) and Humidity (output 2) to their respective gauge nodes:

```javascript
var m = msg.payload.split(',');

// Ensure the data is properly split and contains two values
if (m.length === 2) {
    // Parse the humidity (first value) and temperature (second value)
    var H = { payload: parseFloat(m[0]) };  // Humidity
    var T = { payload: parseFloat(m[1]) };  // Temperature

    // Return humidity and temperature as separate messages
    return [H, T];
} else {
    // If the data doesn't contain two values, return null to indicate a problem
    return null;
}
```

Figure 2: Function Node — JavaScript code to parse humidity and temperature from the serial payload

2.4 Gauge Node Settings

Setting	Temperature Gauge	Humidity Gauge
Title	Temperature	Humidity
Value Format	{{value}} °C	{{value}} %
Min / Max	0 / 50	0 / 100

Step 3 — Deploy and Test

1. Wire the nodes: Serial In → Function → Temperature Gauge (output 1), Humidity Gauge (output 2), Debug.
2. Click the red Deploy button in the top-right corner of Node-RED.
3. Open a browser and navigate to http://127.0.0.1:1880/ui.
4. The live temperature and humidity readings will be displayed as gauge widgets on the dashboard.

Dashboard Output

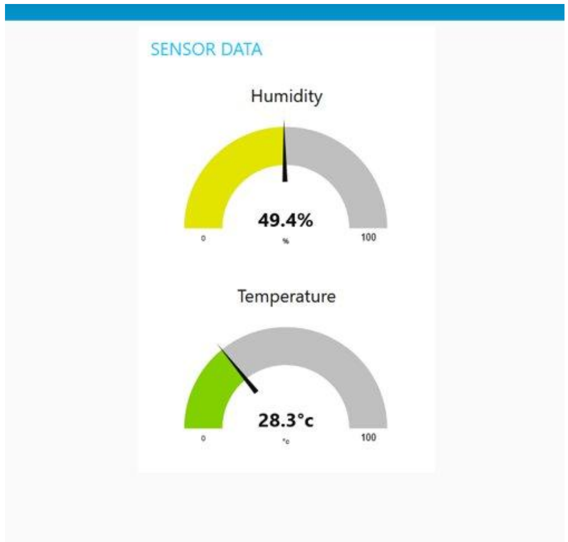

Figure 3: Live Dashboard — Humidity: 49.4% and Temperature: 28.3 °C

Conclusion

This experiment successfully demonstrated the integration of an Arduino Uno R3 with Node-RED for real-time IoT monitoring. The DHT11 sensor transmitted temperature and humidity values via serial communication at 9600 baud. A function node in Node-RED parsed the incoming data and routed each value to a dedicated gauge widget on the web dashboard. The result was a complete IIoT data pipeline — from a physical sensor to a web-based real-time display — built entirely using open-source tools.